KOR STRUCTURAL · ENGINEERING TOOLS

An ETABS model built from the drawings you already have

It removes the typing. It does none of the engineering.

Give it the plan DXFs drafting already exports, and either an ETABS file from the job or just a list of storey names and elevations. It returns a model with the geometry entered — walls at their true thicknesses on their centrelines, columns sized and turned as drawn, floor plates, headers over the openings, shafts and stairs cut out, pier labels so a wall reads as one element up the building.

267 wall panels	645 columns	15 floor plates	14 storeys

31168 YMCA Langara — the YMCA, its podium and the parkade, read from the job's plan sheets. One model per building, at the engineer's request: nothing belonging to either tower is in this file. The ground floor, drafted twice as A-LEVEL 1 and B-LEVEL 1 1.7" apart, is one storey called LEVEL 1. None of it entered by hand.

WHAT YOU GIVE IT

1. The folder of plan DXFs drafting already exports for the job.

2. Somewhere to put it — any ETABS file from that job, even an empty shell with just the storeys. If you have no ETABS file at all, a list of storey names and elevations does.

No configuration. Nothing made specially.

Accuracy rides on two drafting-side things: layer discipline, and sheet titles matching the model's storey names.

WHAT YOU GET BACK

`…-FROM-DRAWINGS.e2k` — import it.

`…-report.txt` — location by location, why anything is missing.

`…-QUESTIONS.xlsx` — every judgement it had to make, most already decided with the measurement behind it. **The rows marked NEEDS YOU are waiting on you**, and the report beside the model counts them at the top: *Questions for you*. Rows tied to a rule can be changed from the answer cell. A second sheet lists every rule the model was built on, read-only.

What it will not touch

No section properties, loads, stiffness modifiers, diaphragms or design — a diaphragm arriving with the geometry is one more thing to undo. And it never overwrites what you have already modelled: on 31138 it recognised 353 of your walls and 391 of your columns and left every one alone.

It gets better every time you tell it something

Every rule came from an engineer and can name the evidence behind it. Rules live in a database with a confidence level, and apply only once verified or confirmed by you.

Both points from your review of 31 August are already in this model. It now arrives turned the way your drawings are — matched to your own grid lines, all 19 on X and both on Y — so the DXF lays straight over it. And *C-LEVEL 3* is the outer edge of the slab at 22,663 sq ft rather than the step inside it: you were right that the outer line is continuous, and it now closes even though two of its segments are drawn on a layer this tool is told is not structure. The stair-stepped edges are gone too — a traced floor is now straightened at the cell it was traced on, so *LEVEL 2* comes to 24 points instead of 114, and an outline that is a staircase stops the publish rather than reaching you.

Override a decision in the workbook	It becomes a rule at the highest confidence there is and applies to every model on every job from then on, superseding ours. You are asked once, ever.
Open a model and say what is wrong	It is measured against your models and the 400-odd on the share, then becomes a rule or a question with the measurement attached. Nothing changes on a hunch. Most recent: "the walls should not break at tower A elevations" — a member now spans its own tower's floor to ceiling in one piece, which took panels under 6 ft from 390 to 63 and columns from 921 to 4.
Ask for something it cannot do	That is where headers as spandrels, pier labels, cut openings and the basement wall came from. Each started as a sentence on a call.

It also learns without being asked: decisions that can be made from a model are recorded as rules with their evidence — including that your 29 spandrels on 31138 imply an 88" opening where an 84" door had been assumed, which had made every header four inches too deep.

How far to trust it

Before it is sent, the model is checked against the drawings both ways — every drawn member must be modelled, and every modelled member must stand on linework from a sheet placed on its own storey. Size and shape are checked against the raw drawing entities, and every check is proved against the defect it exists for. **What it does not know, it says:** on 31168, 7 of its 3,435 drawn members unaccounted for, six storeys whose slab edges will not close, and one roof plate with nothing beneath it. Every judgement it had to make is listed and decided in the workbook with the measurement behind it, so what it assumed is as visible as what it read.
